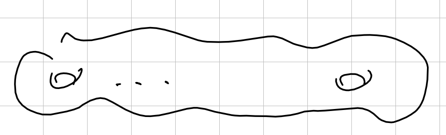
S^n, ~~$S^1\times S^1$~~, $\mathbb{RP}^n$,



(n≥1)
↓
$\pi_1=0$.

2-dim mfd

$L(p,q)$

$D^1\times S^1/e^{2\pi i p}, e^{2\pi i q}$

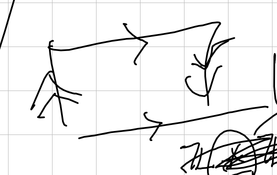
$M \vee N$. ∧ ∨

$[0,1]^2$

torus.

$\mathbb{Z}\times\mathbb{Z}$

Klein.



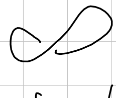
Klein. free product.

~~$\mathbb{Z}\times\mathbb{Z}_2$~~

ababab b.

$\langle a,b \mid aba=b\rangle$.

$aba^{-1}b^{-1}=e$.

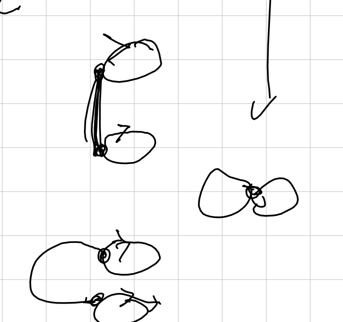


$\langle a,b \mid ab=ba\rangle$

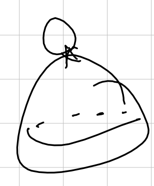
$\mathbb{Z}\times\mathbb{Z}$

ab=ba → grup +Σ.

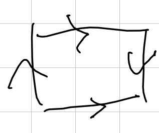
$S^1\vee S^1$

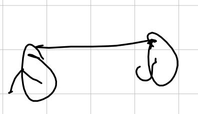


$S^1\vee S^2$



T: $\mathbb{Z}\times\mathbb{Z}$





$S^1\vee S^1$

$\mathbb{Z}*\mathbb{Z}$

$T\setminus\cdot$: $\mathbb{Z}*\mathbb{Z}$

$K\setminus\cdot$: $\mathbb{Z}*\mathbb{Z}$

theorem

S^3 : 0.
↓
$\mathbb{R}^3$: 0.

Theorem

$S^1\vee S^1$

$\mathbb{Z}*\mathbb{Z}$

S^1
↕
$\mathbb{Z}$

M : n-dim
with n≥3

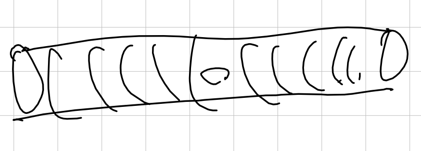
then $\pi_1(M)\simeq\pi_1(M\setminus\cdot)$.

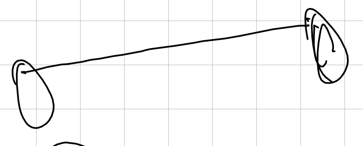
$\simeq\pi_1(M\setminus 2\text{pt})$

$\pi_1(S^n)=0$ $n\geq 2$.

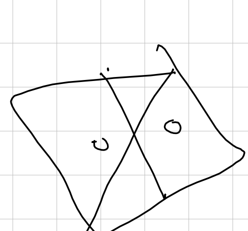
$\pi_1(\mathbb{RP}^n)=\mathbb{Z}_2$ (n≥2).

$\simeq\pi_1(M\setminus\{x_i\})$





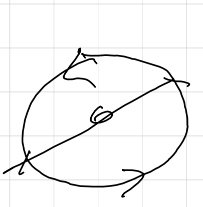
~~tree~~

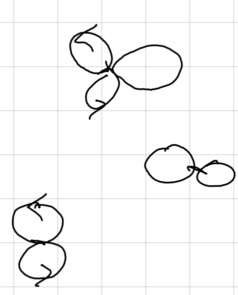


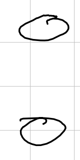
$S^1\vee S^1\vee\mathbb{R}$.
↓
$\mathbb{Z}*\mathbb{Z}$

: $S^1\vee S^1\vee S^1$

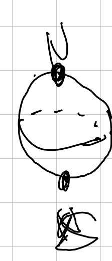
$\mathbb{Z}*\mathbb{Z}*\mathbb{Z}$



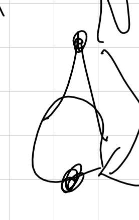


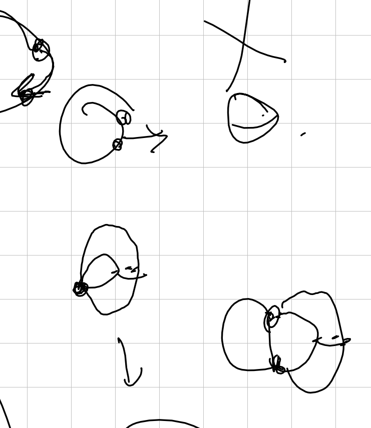


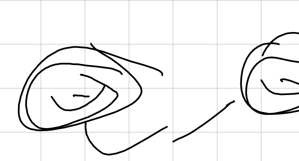
Example.



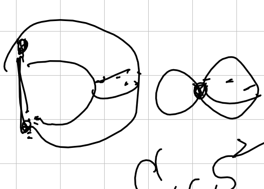
$S^2/N=S$



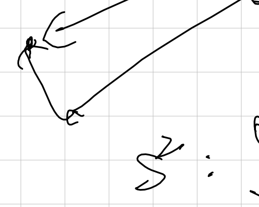


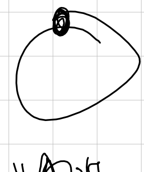


→ $\pi_1(\cdots)$



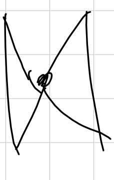
$S^1\vee S^2$





S^1 : 0-cell 1개
1-cell 1개.

S^2 : 0-cell 1개
2-cell 1개



$S^1\vee S^1$

$\mathbb{RP}^2$: 0-cell : 1개
1-cell : 1개
2-cell 1개

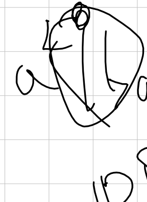
$aba^{-1}b^{-1}=0$.

$\langle a,b \mid ab=ba\rangle$

$(\mathbb{Z}*\mathbb{Z})/\alpha$

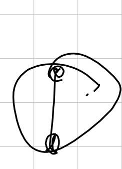
$\mathbb{RP}^2$: $\mathbb{Z}_2$ $\langle a \mid a^2\rangle$

$a^2=1$

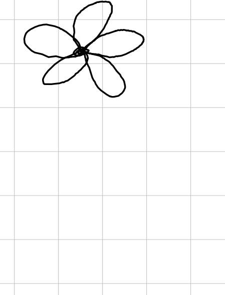


$\mathbb{RP}^3$: ~~0-cell 1개 1-cell 1개 2-cell 1개 3-cell 1개~~

~~$\pi_1(\mathbb{RP}^3)$~~

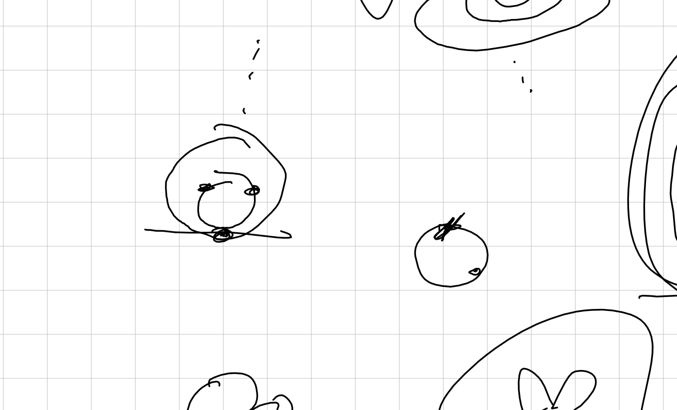


$\pi_1(\mathbb{RP}^3)=\pi_1(\mathbb{RP}^2)=\mathbb{Z}_2$.

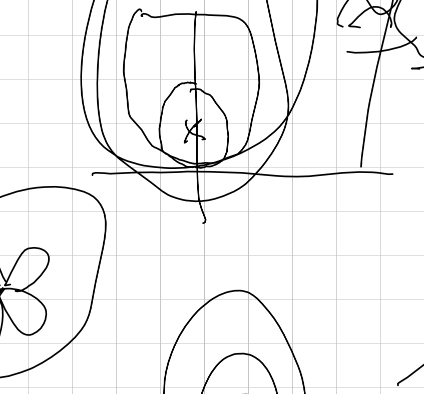


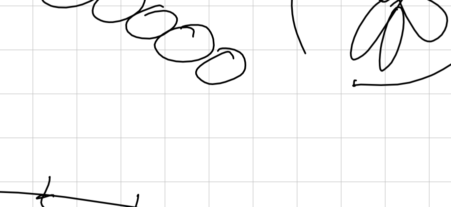
← $x^2+y^2+z^2=1$
∪ $x^2+y^2\leq 1$
∪ $(0,0,z)$ $-1\leq z\leq 1$.

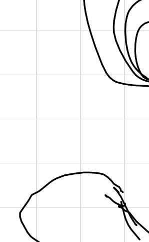
$(\frac{1}{2},\frac{1}{2},\frac{1}{2})$



$\mathbb{R}/\mathbb{Z}$







$P=0$

$Z=x^4+y^4$
↓ exp

$K_1=f(4,2)$
$K_2=f(4)$

$\frac{1}{r^2+z}$.

$\int\frac{Rdz}{y}$ $\int\frac{1}{z}dz$ $\frac{4}{2\pi i}$

$2\pi\cdot X(\cdots)$
0